INCH-POUND

MIL-P-48249A (AR)
AMENDMENT 2
22 May 1991
SUPERSEDING
AMENDMENT 1
27 September 1984

MILITARY SPECIFICATION

PROJECTILE, 155MM, ILLUMINATING, M485A2
ACCESSORIES FOR, AND ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-48249A (AR), dated 30 July 1984 and is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use within all Agencies of the Department of Defense.

* Change Dwg "7549275" to Dwg "8837839" wherever it appears.

PAGE 5

* 3.11, Add the following paragraphs:

"3.11 Control dimensions.

3.11.1 Tool control dimensions. Dimensions marked tool control shall be gaged during First Article Testing and after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor.

* 4.3.1 Add the following first article requirements:

"PART DESCRIPTION	DRAWINGS	QUANTITY
Base Assembly	9224948	10"

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PAGE 9

* Table I First Article Inspection Add the following to sheet 2 of 7:

"Base Assembly
(Dwg 922498)
Examination for defects 10 3.2 4.4.2.35"

PAGE 16

* 4.4.2.1 Shell, illuminant, container.

Delete Minor Defect 201 "Outside diameter .65% 3.2 Gage" and substitute the following:

"201 Inside diameter .65% 3.2 Gage"

PAGE 19

* 4.4.2.3 Major 104: Delete "Position of fins (3 places)" and substitute "position of fins (2 places with reference to third)".

PAGE 20

* 4.4.2.4 Delete Minor Defect 209 in its entirety.

PAGE 21

4.4.2.5 Fin.

Add the following Major Defect:

"Major
101 Thickness .40% 3.2 Gage"

Delete Minor Defects 201 through 209.

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4.4.2.6 Delete Minor Defect 208 in its entirety.

PAGE 23

Delete 4.4.2.7 in its entirety.

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4.4.2.10 Support, parachute, drogue.

Add:

"Major 101	Length		.40%	Gage"

Delete Minor Defects 201 through 205.

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4.4.2.12 Support, parachute, main.

Add:

"Major 102	Length		.40%	3.11	Gage"

Delete Minor Defects 201 through 207.

PAGE 31

4.4.2.15 Delete Minor Defect 204 in its entirety.

PAGE 36

4.4.2.20, Minor: Delete 201 in its entirety.

PAGE 38

* 4.4.2.22, Titles, Change:

From: "Container illuminant."

To: "Candle loading assembly."

* Delete "Major 101; Delete and Substitute:

"Major 101	Outside diameter - shall be press die inside diameter max	Note 1	3.2		Tool Control"

Delete Major 103 closing disc above flush, .40% 3.2, Visual" and substitute in "Major 103 Filling disc above flush, .40% 3.2 Visual".

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Delete "Major 201 closing disc not sealed .65% 3.2 Visual" and substitute "Major 201 Filling disc not secured .65% 3.2 Visual".

* Add note to bottom of page:

"1. Should the Candle Loading Assembly experience an interference fit during assembly into the Canister, Metal Parts Assembly, the consolidation press die shall be dimensionally inspected and the defective dies shall be removed or corrected before production is resumed. The defective parts shall not be included in the lot.

There shall be an approval Recall Calibration System in effect for the consolidation press dies."

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4.4.2.25 Major: Add as follows

"103 Tight pack 0.40 3.2 Manual"

* 4.4.2.25, Minor 201: In column "AQL or 100%" add asterisk symbol "*".

* 4.4.2.25, Minor: Delete 202 in its entirety

* 4.4.2.25, Notes: Add the following: "* Alternate sampling plan may be used as specified in para. 4.4.3.9."

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* 4.4.2.26 Major 103: Delete "Shear pins above flush (4 places)" and substitute "Shear pins above flush or greater than .065" below flush (4 places)."

PAGE 43

* 4.4.2.27 Delete all AQL requirements and requirement paragraph listings of 3.2 and 3.10.

4.4.2.27, Major: Add as follows:

"101 Functional fit with top (9251606) without rotational slip Manual."

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4.4.2.27, Minor: Delete the following Minors in its entirety:

"201, 202, 203, 204."

4.4.2.27, Minor 206: Delete "Gage" and substitute "Visual".

4.4.2.27, Minor 208: Delete "Location" and substitute "Height".

4.4.2.27, Minors: Add as follows:

"211	Length of protrusion	Visual
212	Thickness of material (base)	Gage
213	Bend angle between length and width of protector	Gage".

PAGE 44

* 4.4.2.28 Delete all AQL requirements and requirement paragraph listings of 3.2 and 3.10.

4.4.2.28, Major: Add as follows:

"101	Functional fit with split half (9251605) without rotational slip	Manual"

4.4.2.28, Minor: Delete the following Minors in its entirety:

"201, 202, 203".

PAGE 47

* 4.4.2.31, Critical: Add as follows:

"1 Presence of groove pins (3 each) 100% 3.10 Visual"

PAGE 48

* 4.4.2.32, Major 101: Delete "shear" and substitute "groove".

* 4.4.2.32, Major 105: Delete in its entirety and substitute:

"Lifting plug detorque	.40%	3.2	Gage"

PAGE 50

* 4.4.2.34 Major 103: Delete in its entirety.

Add attached paragraph 4.4.2.35.

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QUALITY CONFORMANCE STANDARD

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.35	Base Assembly	1 OF 1	9224948
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any component missing or improperly assembled	0.40%	3.2	Visual
102	Swivel not free to rotate (360° min)	0.40%	3.2	Manual
Minor				
201	Incomplete crimp resulting in loose disc	0.65%	3.2	Visual
202	Evidence of poor workmanship	0.65%	3.10	Visual

NOTES:

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* 4.4.3.1 Delete "(major defect)".

* 4.4.3.1 Change paragraph 4.4.3.1.3 Firing to 4.4.3.1.6 and add new paragraphs as follows:

"4.4.3.1.3 Continued reduced production sampling. After three consecutive lots have met the criteria of 4.4.3.1.2, 27 projectiles shall be randomly selected for this test. Twenty five (25) of the projectiles shall be fired and observed for compliance with the requirements specified in Table I below. In the event that one or two projectiles show evidence of fuze failure, additional projectile(s) shall be fired to enable the determination of compliance with requirements to be based on a total of 25 projectiles. If a single failure of a projectile to provide "effective illumination" for 75 seconds or more results, an additional 5 projectiles shall be fired to enable the determination of compliance, with the Table I requirements, to be based on a total of 30 projectiles. In the event that two successive lots fail to meet the requirements of 4.4.3.1.3, the sampling provisions of 4.4.3.1.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein, the provisions of 4.4.3.1.2 shall be reinstated.

4.4.3.1.4 Further reduced production sampling. After three consecutive lots have met the criteria of 4.4.3.1.3, 22 projectiles shall be randomly selected for this test. Twenty (20) of the projectiles shall be fired and observed for compliance with the requirements specified in Table I below. In the event that one of two projectiles show evidence of fuze failure, additional projectile(s) shall be fired to enable the determination of compliance with requirements to be based on a total of 20 projectiles. If a single failure of a projectile to provide "effective illumination" for 75 seconds or more results, an additional 10 projectiles shall be fired to enable the determination of compliance with the Table I requirements, to be based on a total of 30 projectiles. In the event that two successive lots fail to meet the requirements of 4.4.3.1.4, the sampling provisions of 4.4.3.1.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein, the provisions of 4.4.3.1.2 shall be reinstated."

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4.4.3.1.5 A stop in production of 10 days or more will require testing in accordance with paragraph 4.4.3.1.1."

Table I, delete in its entirety and substitute the following:

"TABLE I

		SAMPLE SIZE								
		35 or 30			25			20		
		A C C		R E J	A C C	A D D' 5	R E J	A C C	A D D 10	R E J
Category	Defect									
Critical	Premature burst or metal parts separation	0		1	0		1	0		1
Major	Non-Separation with fuze functioning occurring (failure of first ejection)	0		1	0		1	0		1
Major	Projectile does not provide "effective illumination" for 75 seconds or more	1		2	0	1	2	0	1	2
Major	Candle & parachute assembly does not provide for minimum "effective illumination"	See notes 1 & 2								

NOTE 1: The lot shall be rejected if the percent defective "effective illumination" as computed with the use of MIL-STD-414 Section B Table B-3, AQL level 1.5% exceeds the maximum allowable percent defective.

NOTE 2: Only those rounds exhibiting "effective illumination" time of 75 seconds or more shall be considered in computing the percent defective per MIL-STD-414."

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* 4.4.3.9 Add new paragraphs as follows:

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"4.4.3.9 Shear pin holes diameter. The diameter of the shear pin holes shall be inspected in the following manner:

4.4.3.9.1 Sampling. Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every 1/2 hour until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in and of the five (5) units, the machine set up shall be adjusted until (10) consecutive units are defect free.

4.4.3.9.2 Reduced sampling. After successfully completing the sampling of 4.4.3.9.1 for one eight hour production shift with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every one hour and at the beginning and end of the shift. If a defect is found, the sampling plan of 4.4.3.9.1 will reinstated.

4.4.3.9.3 Reduced sampling. After successfully completing the sampling plan of 4.4.3.9.2 for one eight hour shift with no defects, the sampling plan may be reduced to three (3) drilled units being inspected every hour and at the beginning and end of the shift. If a defect is found, the sampling plan of 4.4.3.9.2 will be reinstated.

4.4.3.9.4 Reduced sampling. After successfully completing the sampling plan of 4.4.3.9.3 for one eight hour shift with no defects, the sampling plan may be reduced to two (2) drilled units being inspected every two hours. If a defect is found, the sampling plan of 4.4.3.9.3 will be reinstated.

4.4.3.9.5 Change of tooling or production stoppage. A change of tooling or a stop in production of 60 days or more will require starting the inspection at the first production shift plan."

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4.5.5 Delete "0.100 to 0.125 inch per minute" and substitute "2 inch per minute maximum".

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* 4.5.8 Delete in its entirety and substitute the following paragraphs:

"4.5.8 Proving ground. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The round shall be fitted with an M565 or an M577 fuze. The sample projectile shall be fired with an M4A2 (zone 7) or M119A2 (subject to firing limitation) propelling charge from an M185 cannon. Observation shall be made for conformance to all requirements (see 6.8 and 6.9). The sample projectile shall be tested at the specified zone using weapons, equipment methods and procedures in accordance with the applicable Acceptance Test Procedure. Tubes for this test shall not have less than twenty-five (25) percent remaining tube life based on current gun tube condemnation criteria.

4.5.8.1 Deleted.

4.5.8.2 Safety. The sample projectile shall be fired from an M185 cannon with an M119A2 propelling charge adjusted to give a pressure of 109 - 117 percent of peak pressure at zone 7."

* 4.5.9 Add the following paragraphs:

"4.5.9 Dimensional control of injection molded plastic parts. In place of the normal sampling associated with the Classification of Defects, and after a curing time has been established (see Note 1), (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraph 4.4 after continuous production of each 5,000 parts or at the end of the week, which ever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be inspected for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and AQL of

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0.65 percent or each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request."

PAGE 61

* 6.9.1 Delete the second sentence in its entirety and substitute "All other inadequate or improper functioning rounds under 100 seconds but greater than 75 seconds must be used in calculating the mean."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective the marginal notations and relationship to the last previous amendment.

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